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Roll No.

XCS-600/601(B)

**B. TECH. (SIXTH SEMESTER)
MID SEMESTER
EXAMINATION, March, 2024**

(All Branches)

CAREER SKILLS

Time : 1½ Hours

Maximum Marks : 50

Instructions for students :

1. This paper consists of 50 questions.
2. All questions are compulsory.
3. Each question carries equal marks.
4. Use of calculator is not allowed.
5. There is no negative marking for incorrect responses.
6. It is compulsory to write the set on the OMR.

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Set-B

1. How many 4-digit numbers can be formed using the digits 0 to 6 without repetition such that the number formed is divisible by 5 ?
(a) 240
(b) 200
(c) 220
(d) 120
2. A box contains 5 defective and 15 non-defective bulbs. Two bulbs are chosen at random. What is the probability that both the bulbs are defective ?
(a) $1/20$
(b) $1/19$
(c) $3/19$
(d) $3/20$
3. There are 6 identical Reasoning books and 4 non-identical English books in a library. In how many ways we can arrange the books in a

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shelf (in a row) such that no 2 English books are adjacent to each other ?

- (a) $6! \times {}^7C_4$
(b) $(10!/6!) \times {}^7P_4$
(c) 7C_4
(d) 7P_4
4. Find the probability of getting a sum greater than or equal to 9 in a throw of two dice :
(a) $5/18$
(b) $4/18$
(c) $1/3$
(d) $7/18$
5. What is the probability of selecting a square of dimension (3×3) on a chessboard ?
(a) $2/15$
(b) $1/5$
(c) $4/17$
(d) $3/17$

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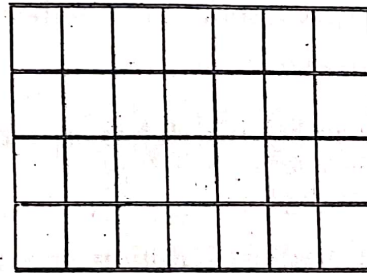
6. 5 boys and 3 girls are to be arranged around a circular table. Find the probability that no 2 girls are together in the arrangement.
- (a) $\frac{3}{7}$
 - (b) $\frac{4}{7}$
 - (c) $\frac{2}{7}$
 - (d) None of the above
7. Find the sum of all the 4-digit numbers that can be formed with the digits 3, 4, 5 and 6 if repetition of digits is not allowed :
- (a) 119988
 - (b) 11988
 - (c) 191988
 - (d) None of the above
8. There are 12 points in a plane out of which 5 are collinear. Find the number of unique straight lines that can be drawn by using these points :
- (a) 66
 - (b) 56
 - (c) 57
 - (d) 67

9. In a tournament each team played against every other team once. The total number of matches played is 45. Which among the following is true about the number of teams in the tournament ?
- (a) It is a number which has an odd number of factors.
 - (b) It is the smallest 2-digit prime number.
 - (c) It is a number that has 2 prime factors.
 - (d) It is a number that has a total of 6 factors.
10. Find the number of positive integral solutions to the equation $a + b + c + d = 12$.
- (a) 455
 - (b) 165
 - (c) 305
 - (d) None of the above
11. There is a network of horizontal and vertical roads as shown in the figure below. A person has to go from point P to point Q taking the

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shortest path. Find the probability that the first two paths taken by the person are both horizontal :



- (a) $21/55$
 - (b) $27/55$
 - (c) $34/55$
 - (d) None of the above
12. 3 cards are drawn at random from a pack of cards. Find the probability that the cards are of both colours.
- (a) $14/17$
 - (b) $13/17$
 - (c) $12/17$
 - (d) $11/17$

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13. All possible arrangements of the letters of the word DIFFER is being done as in a dictionary (meaning is not important). Find the word which will have the 50th rank :

- (a) DREFFI
- (b) DRFEFI
- (c) DRIEFF
- (d) DREFIF

14. How many 5 digit numbers can be formed using the digits 1, 2, 3, 4, 5 and 6 without repetition such that the even places can only be occupied by the even digits ?

- (a) 150 ways
- (b) 172 ways
- (c) 144 ways
- (d) None of the above

15. The odds in favour of India winning the T20 World Cup is 5 : 3 while the odds against Australia winning the T20 World Cup is 3 : 4.

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Find the probability that either India or Australia wins the T20 World Cup.

- (a) $1/2$
- (b) $47/56$
- (c) $25/56$
- (d) $2/3$

16. The floors in a seven floor building are numbered 1 to 7. A person gets in a lift on 3rd floor. What is the probability that the person gets out of the lift on a floor which is a prime number? (He may go out on the 3rd floor also)

- (a) $5/7$
- (b) $4/7$
- (c) $3/7$
- (d) None of the above

17. In a bag contains 9 red and 6 green balls. What is the probability that 3 balls are red and 2 are green if 5 balls are drawn at random?

- (a) $8/19$
- (b) $19/41$
- (c) $41/60$
- (d) $60/143$

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18. A candidate is required to answer 7 questions out of 12 questions, which are divided into two groups, each containing 6 questions. He is not permitted to attempt more than 5 from either group. In how many different ways can he choose the seven questions?

- (a) 780
- (b) 640
- (c) 920
- (d) 1020

19. In how many ways can a team of 3 students be selected out of 4 boys and 3 girls such that the team has at least 1 boy?

- (a) 34 ways
- (b) 35 ways
- (c) 4 ways
- (d) 12 ways

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20. How many 4-digit numbers not greater than 3000 can be formed using the digits 0 to 5 if repetition of digits is allowed ?
- (a) 512
(b) 1296
(c) 648
(d) 513
21. In how many ways can the alphabets of the word "Instantaneous" be arranged such that the vowels are always together ?
- (a) $13!$
(b) $13!/(3! \times 2! \times 2! \times 2!)$
(c) $(8! \times 6!)/(3! \times 2! \times 2! \times 2!)$
(d) $(8! \times 7!)/(3! \times 2! \times 2! \times 2!)$
22. The probability of 3 people failing in an exam is $2/3$, $1/4$ and $3/5$ respectively. Find the probability that exactly 2 of them pass the exam :
- (a) $5/12$
(b) $23/60$
(c) $7/12$
(d) $2/3$

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23. An integer is chosen at random from the first 100 natural numbers Find the probability that it is neither divisible by 5 nor by 8 :
- (a) $3/10$
(b) $7/10$
(c) $9/10$
(d) $4/5$
24. A task is assigned by team leader to 2 members Anthony and Iqbal separately. The probability that Anthony will complete the task is $2/3$ and the probability that Iqbal will complete the task is $3/4$. What is the probability that the task will be completed ?
- (a) $11/12$
(b) $10/11$
(c) $8/9$
(d) $7/8$

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25. Aakash and Anant are playing chess. The probability that Aakash wins a match between him and Anant is 0.8. Both played 3 games which are independent to each other. What is the probability that Aakash will win exactly 2 games?

- (a) 63/125
- (b) 53/125
- (c) 48/125
- (d) 87/125

Directions (Q. Nos. 26-29) : Applying the principle of parallelism, find out which of the following options will replace the blank and mark that option as your answer.

26. After Mrs. Gupta cashes her paycheck, the money goes to her savings account, cappuccino fund, and _____.

- (a) paying her credit card bill

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- (b) toward her credit card balance
- (c) credit card balance
- (d) towards her credit card balance.

27. Not only did Maria despise the way June chewed with her mouth open, _____ finding her wet towels all over the bathroom floor.

- (a) but also disliked
- (b) but also disliking
- (c) but he also disliked
- (d) but he had also disliked

28. To protect her delicate hands, Sam will not rake the yard, do the dishes, or _____.

- (a) refuses to scrub the bathtub
- (b) won't scrub the bathtub
- (c) scrub the bathtub
- (d) will not scrub the bathtub

29. In Mrs. Cinthia's office, Raoul feigned the flu. He hoped to be excused from class, get an extension on his paper, and _____.

- (a) spend the afternoon playing tennis with his friends

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- (b) to spend the afternoon playing tennis with his friends
- (c) then to spend the afternoon playing tennis with his friends
- (d) spent the afternoon playing tennis with his friends

Directions (Q. Nos. 30-33) : Read the sentences carefully and find out which part has an error and indicate that part as your option.

- 30. I am working at my present job (a) / since the day (b) / my younger sister was born. (c) / No error (d)
- 31. If Mamta was able to create jobs (a) / and reverse the exodus of talent from Bengal (b) / she will truly set the deal (c) / on her legacy as Bengal's savior. (d)
- 32. The boss reminded him (a) / of the saying that honesty was the best policy (b) / and told them (c) / that they better be honest in their work. (d)
- 33. I'll try to get there (a) / at 6.30 unless you want me (b) / to arrive earlier (c) / No error (d)

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Directions (Q. Nos. 34-38) : In each of the given sentences, a part is underlined. Replace the underlined part with the correct option and mark that as your answer.

- 34. Reporting a 90 percent drop in net income during the second quarter, dragged down by restructuring charges and weak sales, the earnings guidance for the year was withdrawn by Best Buy Co.
 - (a) Report of a 90 percent drop in net income.
 - (b) Best Buy Co. is reporting a 90 percent drop in net income.
 - (c) Best Buy Co. has reported a 90 percent drop in net income.
 - (d) Reporting of a 90 percent drop in net income.
- 35. Recently, hopes that the European Central Bank will reach a deal to help Spain and Italy borrow at cheaper rates has nudged financial markets higher.
 - (a) Spain and Italy borrow at cheaper rates has nudged financial markets higher.

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(b) Spain and Italy borrow at cheaper rates have nudged financial markets to higher levels.

(c) Spain and Italy borrow at cheaper rates has nudged financial markets to higher levels.

(d) Spain and Italy borrow at cheaper rates nudged financial markets to higher levels.

36. The event of the migration of European people have been adequately recorded in the histories of the Middle Eastern regions.

(a) have been adequately recorded in the histories.

(b) has been adequately recorded in the histories.

(c) have been adequately recorded in the history.

(d) has been adequately recorded in the history.

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37. The sense of elitism in the art world is deepened by the fact that the determination of the authenticity of long-lost works of reputed renaissance-era painters such as Da Vinci is usually not possible by just scientific analysis and forensics but need the expert "eye" of the connoisseurs.

(a) painters such as Da Vinci is usually not possible by just scientific analysis and forensics but need.

(b) painters like Da Vinci is usually not possible by just scientific analysis and forensics but needs.

(c) painters like Da Vinci are usually not possible by just scientific analysis and forensics but needs.

(d) painters such as Da Vinci is usually not possible by just scientific analysis and forensics but needs.

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38. In the U.S., German luxury brands have succeeded in getting people who might have once thought the cars were out of their League in buying and leasing some snazzy rides.

- (a) have succeeded in getting people who might have once thought the cars were out of their League in buying and leasing some snazzy rides.
- (b) have succeeded in getting people who might have once thought the cars were out of their League to buy and lease some snazzy rides.
- (c) have succeeded in getting people who once might have thought the cars were out of their league to buy and lease some snazzy rides.
- (d) has succeeded in getting people who might have once thought the cars were out their league to buy and lease some snazzy rides.

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Directions (Q. Nos. 39-50) : Identify which part of the sentence has an error and mark that part as your answer. In case of no error, mark option (d) as your answer.

- 39. A new study has shown (a) / that children who have high body mass index (b) / are more likelier to have high blood pressure, cholesterol and blood insulin levels by the time they reach adolescence. (c) / No error (d)
- 40. The success of the government sponsor (a) / job guarantee program has resulted (b) / in a drastic drop in poverty. (c) / No error (d)
- 41. Consumers, these days may be led (a) / to spend ridiculous amounts of money on products (b) / that supposedly restore his or her youth and beauty. (c) / No error (d)
- 42. The Indian team cannot win the series (a) / and even pulling off a draw could be (b) / a tough proposition with its lead bowlers not playing. (c) / No error (d)
- 43. The rate of metabolism of a body is (a) / comparatively lower (b) / when it is at rest and is thus optimum for examination. (c) / No error (d)

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